# Khaled Abdelhadi

**MECHANICAL ENGINEER · THERMAL & FLUID SIMULATION · DESIGN & FEM · HYDRAULICS · MANUFACTURING**

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Interactive CV: k95-de.github.io/cv | **German citizen** | Languages: **German** fluent · **English** fluent · **Arabic** native

## 01 PROFILE

Mechanical engineer (M.Eng.) with **5 years of thermal management, hydraulics and 1D/3D simulation** at MAN Truck & Bus and TRATON R&D. I build and validate **cooling-system, hydraulic and heat-transfer models**, run **pressure-loss and pipe-sizing calculations**, correlate results against CFD and test data, and turn them into **design decisions**: component sizing, pipe routing, pump operating points and failure prevention. Hands-on in **3D part and assembly design, FEM-based structural optimization, composites and mold manufacturing**.

Thermal Management | 1D CFD / GT-SUITE | Hydraulics | Pressure-Loss & Pipe Sizing | FEM / FEA | Stress & Stiffness | Structural Optimization | 3D CAD Design | 2D Drawings & Tolerances | Failure Analysis | Test Correlation | Manufacturing | Python · MATLAB

## 02 EXPERTISE

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|---|---|
| **Thermal** | Thermal management, heat transfer, thermodynamics, cooling circuits (battery, e-motor, HV, cabin, refrigerant), heat exchangers, transient drive-cycle simulation, extreme-ambient, worst-case and safety (FTTI) scenarios, thermal hotspot analysis |
| **Hydraulic & Fluid** | Hydraulic circuit modelling, pressure loss and pipe sizing calculations, flow rate and flow distribution analysis, pump sizing and operating points, venting and restrictor design, hydraulic validation against measurements, P&ID reading, industrial piping (ASME B31.1 / B31.3 / B36, single- and two-phase pipe hydraulics, valves, flanges) |
| **Simulation** | 1D CFD / system simulation (GT-SUITE, expert), 3D CFD correlation, FEM / FEA structural simulation (Altair HyperWorks; Abaqus / Nastran basic), electromagnetic simulation (ANSYS Maxwell), plastic-flow simulation, model calibration and validation against test data, simulation methodology |
| **Design & FEM** | 3D part and assembly design, 2D drawings with tolerances (DIN / ISO), mechanical design of components, stress and stiffness assessment, structural optimization (weight, torsional stiffness), CFRP composites, BOM creation; CAD: PTC Creo, SpaceClaim, Revit (BIM) |
| **Manufacturing** | Injection mold (steel tool) design, CNC programming, composite layup and curing, prototype build, supplier data evaluation, molding-defect elimination, manufacturing documentation |
| **Tools & Methods** | MATLAB/Simulink, Python (data analysis, automation), AUDI loss-map toolbox, JIRA, Agile / Scrum, V-model / ASPICE, failure analysis and root-cause investigation, trade-off studies, KPI tracking, technical documentation |

## 03 PROFESSIONAL EXPERIENCE

### 1D Simulation Expert, Vehicle Cooling Systems — Apr 2025 – Feb 2026
TRATON R&D Germany GmbH, Nürnberg

### 1D Simulation Engineer, Vehicle Cooling Systems — May 2022 – Apr 2025
MAN Truck & Bus SE, Nürnberg

GT-SUITE · 3D CFD correlation · test-bench data · JIRA / Agile

- Led thermal-management simulation for electric and diesel truck cooling systems; owner of the 1D cooling-circuit models used for concept decisions and release.
- Built 1D hydraulic and thermal models from 3D pipe geometry and supplier data; calibrated coolers and circuits against 3D CFD and physical measurements with strong correlation.
- Performed pressure loss, flow rate, flow distribution and pipe sizing calculations across interconnected battery, HV-component, cabin and refrigerant circuits; optimized pump sizing and operating points, improving energy efficiency.
- Ran worst-case, extreme-ambient and safety scenarios (incl. FTTI) and derived design changes: venting positions, flow restrictors, pipe diameters and elevations.
- Performed failure analysis on cooling issues and supported component selection with simulation-based trade-off studies for suppliers and design teams.

- Created configurable variant sub-models (e.g. battery count) reused across vehicle platforms; wrote the team's simulation methodology, cutting onboarding time and standardizing model quality.

### 1D/2D Simulation Engineer, E-Motor Cooling
Dec 2021 – May 2022

MAN Truck & Bus SE, Nürnberg

ANSYS Maxwell · AUDI loss-map toolbox · GT-SUITE

- Simulated e-motor electromagnetic behaviour and losses; validated torque/speed against measurements and generated loss maps for the thermal model.
- Calibrated the GT thermal model to sensor data, achieving temperature predictions within measurement tolerance.
- Identified thermal hotspots, demagnetization risk and previously unaccounted system losses before hardware testing.

### 1D Simulation Student, E-Motor Cooling
Oct 2020 – Mar 2021

MAN Truck & Bus SE, Nürnberg

- Built a bus e-motor 1D thermal model in GT-SUITE from scratch, including oil-circuit design with conduction and convection paths; became the base model for all later e-motor cooling studies.
- Converted 3D geometry to 1D and validated transient thermal simulation against drive-cycle test data.

### Structural Simulation Engineer, CFRP Monocoque (Formula Student)
Mar 2020 – Aug 2020

Esslingen Rennstall, Esslingen

Altair HyperWorks · FEM · CFRP

- Optimized the CFRP monocoque of the Stallardo 19 race car with FEM for lower weight and higher torsional stiffness against structural requirements.
- Delivered the production-ready BOM and 2D schematics for the Stallardo 20 chassis, translating simulation results into manufacturing documentation.

### Engineering Development & Management Intern
Jan 2020 – Aug 2020

A123 Systems GmbH, Stuttgart

- Built project KPIs from JIRA tickets across V-model (ASPICE) phases, enabling data-driven milestone forecasting and improving team velocity.
- Analysed BMS hardware/software architecture, mechanical battery design and safety-testing protocols.

### Mold Design & Manufacturing Intern
Jul 2017 – Jan 2018

Alfanar Electric, Riyadh

PTC Creo · CNC · plastic-flow simulation

- Designed steel injection molds as 3D assemblies with 2D drawings, programmed CNC machining and ran plastic-flow simulations to identify and eliminate molding defects.

### CFRP Design & Manufacturing Member
Jun 2016 – Jun 2018

Shell Eco-Marathon Asia, Singapore

- Designed the vehicle exterior in Creo, built CNC-machined foam molds, led carbon-fibre layup and curing; monocoque validated through tensile testing.

## 04 EDUCATION

**M.Eng. Design & Development in Mechanical & Automotive Engineering**
Hochschule Esslingen, Germany

**B.Sc. Mechanical Engineering**
Alfaisal University, Riyadh, Saudi Arabia

## 05 CERTIFICATIONS & TRAINING

- **Design of Industrial Piping Systems** — L&T EduTech / Coursera, 2026 (pipe hydraulics, pipe sizing, ASME B31, pipeline construction)
- **AI for Design and Optimization** — University of Michigan, 2026 · **Python for Data Science, AI & Development** — IBM, 2026
- **Computational Fluid Mechanics** (project) — Coursera, 2026 · **BIM Fundamentals for Engineers** — L&T EduTech, 2026
- **Claude Code 101** — course completion badge, Claude Academy (Anthropic), Aug 2026 · verify · **SimOps Fundamentals** — 2025
- **ITIL 4 Foundation & Digital Business Management** — Habmann Group, in progress until Dec 2026